2026 FALL

SE_Fall2026 – Project Deliverable (D3)

The objective of the Specification Deliverable (D2) is to outline the *system requirements of the component – game platform or game engine - your team has built* during the Sprint 1 and Sprint 2 of this course.

This document will be released to the rest of the class by the end of the sprint, once all the documents are submitted. As the D3 will be used to advertise your component for the second phase of the project, you want it to be as complete and clear as possible.

To help you understand what it is expected in the D3, below you can find a detailed outline of all the sections you can provide, with some examples of the content expected.

NOTE: Remember to add somewhere in this document the contact information for your group!

1. Scope statement

The **scope statement** section of the requirements should briefly describe the purpose of the current document and to summarize its content in natural language. While doing so, you should introduce the component you have developed and to highlight the main requirements that could possibly differentiate your component from the components from another teams.

For example, the scope statement of a D3 document for an Anti-cyberbullying System could be this:

This document outlines the requirements for an Anti-cyberbullying System (ACS) that is used by moderators in order to monitor an immersive game environment (IGE), where players from different schools can interact with one another. The system should be able to detect certain actions, defined as cyberbullying flags (CBF) in this document. The flag detection should be automated, but there should also be a way for the players to manually ag something as bullying.

Based on in-game action log les sent by the IGE, the ACS system should detect CBFs and present them to the responsible moderators. All the action logs related to a CBF, should be available through an interface. This interface should also allow the management and configuration of those flags. Moderators should be able to take direct action with regard to flags and generate cyberbullying events (CBE), when needed. Apart from monitoring the IGE, they should also be able to interact in the game as master players.

The recommended size of this section is between 1 and 3 paragraphs.

2. Definitions

This section of the requirements is mandatory. The main purpose of the section is to provide **definitions** for any relevant term, technical or not, necessary to understand the rest of the D3.

In the context of the Anti-cyberbullying System it could be useful, for example, to clarify what the customers considered "cyberbullying" or a "cyberbullying event":

2.1 Cyberbullying

Cyberbullying is the act of bullying through electronic media, such as social media, IGEs, e-mails, cell phones and more. There are different types of cyberbullying, some of them are discussed below (G. and M. 2015).

1. *Flaming*, the use of vulgar and angry language when communicating.
2. *Harassment*, repeated unwanted messages or stalking where noticeable (such as in an IGE).
3. *Denigration*, the act of disrespecting someone, or libel/defamation.
4. [⋯]

2.2 Cyberbullying Event

[⋯] Different violations may have different flags associated with them. Hence, any action that raises a flag is referred to as a "flag violation", if this action is confirmed as an act of bullying it will be known as a Cyberbullying Event (CBE).

There is no recommended size for this section but try to be as thorough as possible to make possible to understand the content of the following sections. Each new concept of your solution (e.g., piece, board, player, server, tournament, file, file format) must be properly defined.

You will also need to define the rules of the game or the rules for the tournament somewhere. This is the recommended place for it!

3. Components

The **components** section of the system requirements should give the necessary context to understand the functional and non-functional requirements and your component's architecture. The main content of the section is a *simplified component diagram,* plus ore or one *component diagrams* of your *system architecture* with a brief description of each one of the components that integrates the solution.

We know that at this point your team knows little or nothing about the other components of the UU-GAME, but it is always useful to write down how you imagine the several systems interact.

To write this section,

1. Ignore the physical architecture of the application, then decide what will be the main subsystems of the project. Draw a simplified component diagram depicting these logical relations.
2. Then, proceed adding the details: add the input and output interfaces and split the original components if needed.

In the context of the Anti-cyberbullying System this section could look like (***VT 2021: as requested by students in previous editions we provide a larger example this year, hoping to get a better D3***).

3 System architecture

Due to the demands of real-time intervention, the architecture must enable the system to process data as fast as possible, but it must also allow the moderator to intervene within the IGE to prevent a situation that might escalate. Therefore there are two possible system architectures, the pipe and filter, or the client-server architecture. The client-server architecture makes the system a lot more manageable and easier to parallelize. The pipe and filter architecture, on the other hand, is an easier system to build but is more likely to run in to performance and maintain- ability issues. Because of this, the recommended architecture for this system is a client-server architecture, using a pipe and filter architecture within the server as follows.

3.1 Server architecture

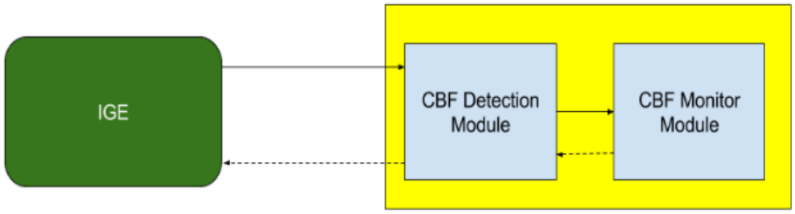


Figure 1 Pipe and Filter Based Server

These three components will work together to form the server of the ACS. Because of the performance focused nature of these components the recommended system architecture for this server is a pipe-and-filter architecture, allowing it to quickly work through a lot of log files.

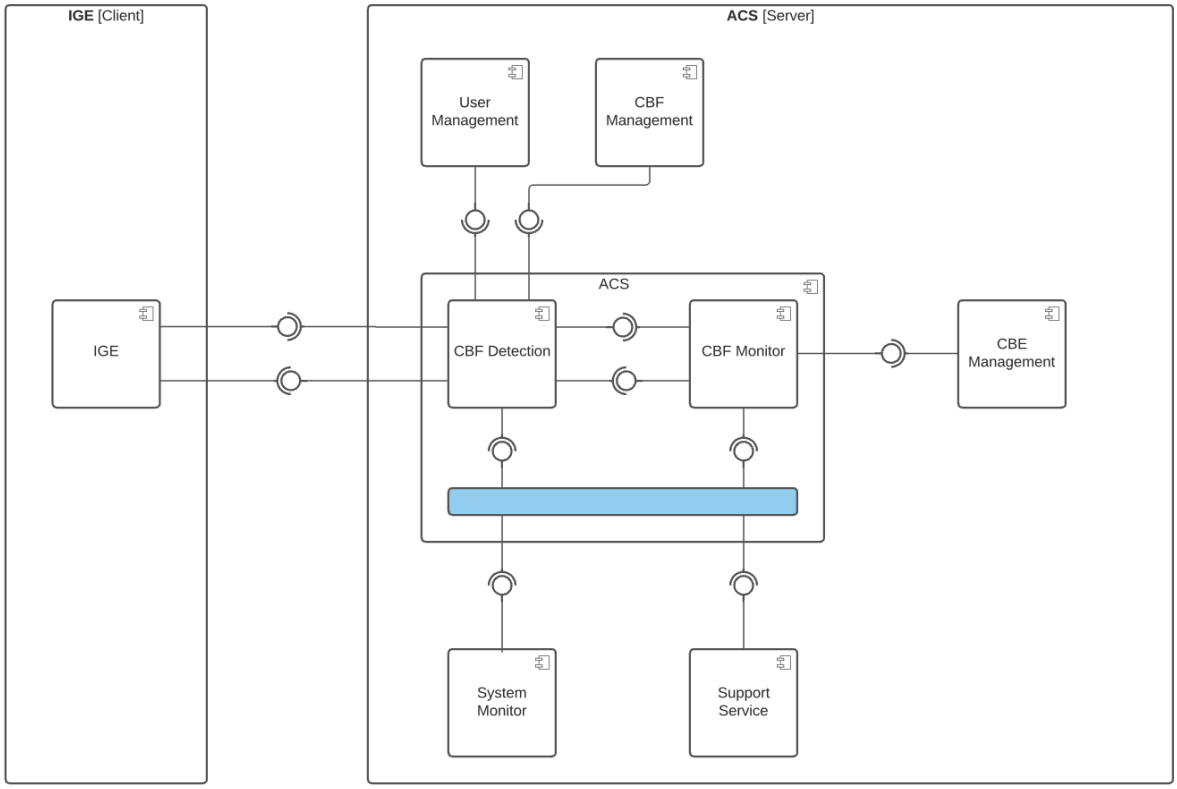


Figure 2 Complete System Architecture

The client systems are composed of the different modules used to edit options that the server uses to filter logs. As they're not heavily interconnected this allows multiple users to access the same parts of the system, which must be controlled, perhaps using a mutex lock or a semaphore type system. The IGE and SIS will also be clients in this system architecture, merely feeding the ACS Server with information or data to allow it to function.

The description of each sub-component should not be larger than one (1) paragraph (4-5 lines). We have discussed Figure 2 in detail in Lecture 4 Architectural Design.

4. Functional requirements

The **functional requirements** section provides the functional requirements specification for the developers. Remember that they are not a complete description of the behavior of your system, but a brief summary of the requirements and the rationale behind it. This information will be used by other groups to understand why the requirement is needed and to track the requirements during the outsourcing exercise.

The *functional requirements* must focus on the behavior of the system rather than how the users interact with them. You only need to provide the requirements for the component you have built.

Back to the Anti-cyberbullying System, a user requirement for the “admin” was:

REQ-ADMIN2. The administrator needs to be able to create pre-defined CBF templates, so moderators can choose one or another depending on the type of forum.

One of the corresponding functional system requirements can be:

ID and Name:	REQ-ADMIN 2.1 Create CBF templates
Created by:	Teachers
Date created:	2023-08-21
Primary actor:	Administrator
Secondary actors:	Moderators
Description:	The templates selection system shall show the list of pre-defined CBF templates to any administrator logged into the system as a table with the template name, the template description, and the name of the creator. An administrator shall be able to create a new pre-defined CBF template using “Upload Template” form.
Trigger:	An administrator clicks on the “New Template” button
Preconditions:	PRE-ADMIN 2.1-1 The user has been authenticated into the system. PRE-ADMIN 2.1-2 The user has the “administrator” role. PRE-ADMIN 2.1-3 The user is in the template management system.
Postconditions:	POST-ADMIN 2.1-1 The new template is saved in the database. POST-ADMIN 2.1-1 The system shall show the list of pre-defined CBF templates.
Normal flow:	**REQ-ADMIN 2.1a Create a new CBF Template** 1. The administrator specifies a new name for the template in alphanumeric. 2. If they want, the administrator specifics a description for the template. 3. System requests the Database Manager to store the new CBF template. 4. All CBF templates views are asynchronously updated, so moderators can observe the new template. 5. System returns to the previous view, listing all pre-defined CBF templates.
Alternative flows:	<none>
Exceptions:	**REQ-ADMIN 2.1/EXCEPTION 1. The template name is duplicated** 1 The system displays message: “Duplicated template name.” 2 System asks administrator if they wants to update existing template (3a), choose a different name (4a) or exit (5a). 3a. Administrator asks to update existing template. 3b. System requests to the Database Manager to make a temporal copy of the template. 3c. System starts normal flow over using the temporal copy. 3d. When the normal flow finishes, the system requests to the Database Manager to override the original copy. 4a. Administrator asks to choose a different name. 4b. Restart the functional requirement. 5a. Admin asks to exit. 5b. System terminates functional requirement. **REQ-ADMIN 2.1/EXCEPTION 2. The template name uses non-alphanumerical symbols.** 1 The system displays message: “Invalid template name. You can only use alphanumeric values.” 2. System starts normal flow over.
Diagrams	Figure 1
User requirement:	REQ-ADMIN2

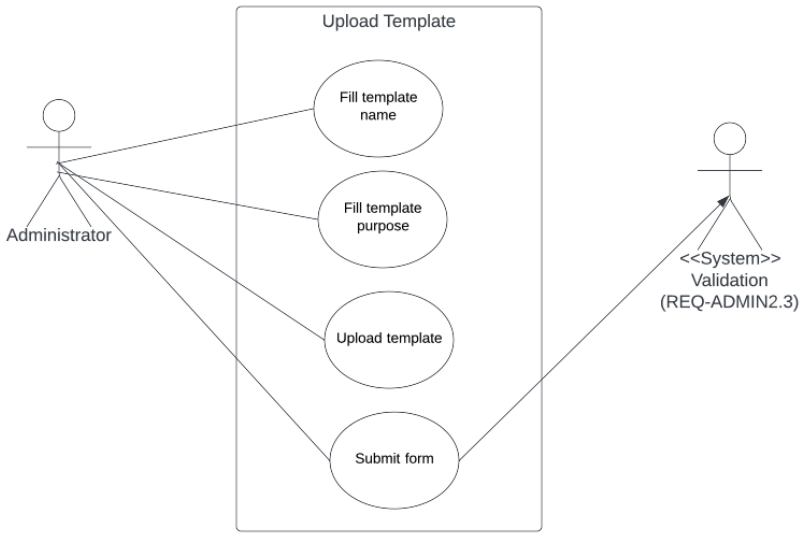


Figure 1: Administrator interaction diagram

You can find another example of a system requirements in the slides of Lecture 2.

5. Non-function requirements

The **non-functional requirements** section of the requirements should briefly describe the limitations of the solution imposed by some externality. Unlike the system constrains, these are enforced and must be followed. In fact, remember that most of the non-functional requirements must be aligned with the architectural design of your solution (Section 3).

The best way to describe this section is to add a small subsection for every one of the type of non-functional requirements. We could have many examples from the Anti-cyberbullying System:

6.1 Space requirements

Thee ACS should be able to store a large amount of recorded data and be able to read and manipulate it efficiently and without causing performance problems for the rest of the system.

6.2 Reliability requirements

Redundant storage is required in order to ensure the availability of the ACS. Downtime of the system shall not exceed 1% of the office hours. Maintenance processes should be undertaken in late night according to the geographic location of the server. e ACS shall ensure recovery stages for critical processes such as in-game action log les, CBF reports and warnings.

A. Changelog

If the document is not the first version, provide a list of the changes that have been applied since the last deliverable. For example, Team Z has delivered a 3rd version of the document:

VERSION	DATE	CHANGE
V1	2022-10-05	Initial version
V2	2022-10-17	Added REQ-ADMIN-3 Added REQ-ADMIN-4 Added REQ-ADMIN-7 Deleted REQ-ADMIN-14 Modified REQ-USER-5 Modified REQ-USER-6 Improved description of USER Added description of ADMIN
V3		

B. Grading – Individual Contributions

Each one of the members of the group **must** be responsible of writing at least **2** of the functional requirements (Section 4) and **3** of the non-functional requirements (Section 5) of this document. The quality of these system requirements will be part of the individual grade of the student. The grade of other students will not be affected (positively or negatively) by this part of the assignment.

Please, indicate who has written each one of the user requirements in the following table:

NAME	INDIVIDUAL CONTRIBUTIONS
SHAHID MANOOR	REQ-ADMIN-2.1 REQ-ADMIN-2.3 Non-Functional Requirement 6.2, 6.5 and 6.6
MUHAMMAD ALI	…
KHURRAM SHAHZAD	…